

(TR-104) PYTHON WITH AI/ML

Training Day 57 Report

Date: 31 March 2026

Topic Covered: AI Agents & Tool Calling (Structured Outputs and API Integration)

The fifty-seventh day of training focused on understanding AI agents, their working mechanism, and practical implementation of tool calling using LLMs. The session included hands-on experimentation with structured outputs, API integration, and building basic agent-like pipelines.

- Learned the fundamental concept of AI agents and how they differ from traditional chatbots.
- Understood how AI agents perform actions instead of only generating responses.
- Explored how agents break tasks into step-by-step execution flows.
- Learned how AI agents interact with external tools and APIs.
- Practiced structured output generation using JSON formatting for reliable agent responses.
- Built simple agent-like pipelines using Ollama models for experimentation.
- Implemented city extraction from user queries using LLM-based parsing.
- Integrated weather API using geocoding services and Open-Meteo API for real-time data retrieval.
- Handled JSON parsing and corrected invalid structured outputs from models.
- Managed API errors and improved robustness of tool-calling workflows.
- Strengthened understanding of function calling and tool integration in modern AI systems.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/agents-practice>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/turning-prompts-into-actions-my-first-ai-agent-with-ollama-weather-api-9fea017dd6b9>